

The green transition and economic polarization in Europe: a multi-level assessment with Germany and Poland as case studies

Detailed project description

1. Scientific goal of the project

1.1. Scientific goal of the project, description of the problem to be solved

The main goal of the proposed project is to study **how the policy measures of the European green transition affect regions and different Member States**, and whether heterogeneous impacts bear the **risk of fuelling a socioeconomic polarization** within the EU. Based on these insights, the project will delineate implications for the “strong policy response at all levels” that the European Commission considers to be necessary (COM/2019/640, sec. 2.2.1).

To achieve this goal, the project is structured into **three work packages**, each of which is focused on a different level of analysis, as well as **a final package meant to synthesize the findings**:

1. The first work package delineates a **national (macro) country topology** based on the vulnerabilities and potentials of the EU economies regarding a green transition; the typology helps identifying **macro constraints for regional policies and implications for comparative economic development in Europe**.
2. The second work package focuses on the **regional (meso) level** of European coal regions, which will be especially affected by a green transition. The work package comprises a quantitative analysis of **regional-level preconditions towards green energy transition**, especially the possibility of the regions to diversity into technologically related economic activities and their geographical pre-conditions.
3. The third work package complements the quantitative work of the first two work packages by **two representative case studies on the local (micro) level**; it comprises a comparative analysis of two lignite regions - Konin (Poland) and Lusatia (Germany) – employing a **qualitative analysis** of factors influencing the success of the ‘just transition’.
4. The fourth work package is dedicated to the **identification of linkages between the three levels of analysis** and the **delineation of policy implications** based on the integrated results of the first three work packages: it discusses to what extent the green transition should be accompanied by measures that prevent a growing polarization among Member States, and that secure the political acceptance of the reforms that are needed for Europe to meet its sustainability goals.

The project contributes to the ongoing academic and policy debate on the **effects of the increased commitments of the EU to environmental-friendly development**, as evidenced in the ratification of the Paris Agreement in 2016, its operationalization in the Katowice Agreement in 2018, the proposal of the European Green Deal in 2020 and EU position at the COP26 in Glasgow in 2021. To comply with these obligations, and to put EU economies on a 1.5 degree-compatible emission reduction pathway, an ecological transformation of the production and consumption sphere of the European Union – here summarized under the heading ‘green transition’ – is inevitable. This involves deep structural reforms towards more environmentally friendly production structures and the provision of incentives that foster more sustainable consumption patterns. In the EU there are currently **21 coal regions** (on NUTS-2 level), with coal mining providing directly about 185,000 jobs in total in the European Union and additionally 53,000 in coal-fired power plants. Poland and Germany hold the largest number of jobs in coal mines (100,000 in Poland, 25,000 in Germany) and coal-fired power plants (13,000 in Poland, 11,000 in Germany), as well as indirect jobs in coal-related activities (136,506 in Poland and 48,455 in Germany, Alves Dias et al. 2018). A green transformation is (or is at least presented as) a potential source of instability, threatening important sources of value added and employment in these regions.

Among **diverse potential socioeconomic effects** of the planned energy transition on the regional level, the risk of economic polarization stands out in the literature as a major threat, as well as a potential political barrier for the necessary reforms. For some of the EU member states, as well as regions and social actors, **the transition appears as a challenge** which may undermine their economic status. For others, meanwhile, it is **perceived as a window of opportunity**. The distribution of risks and challenges associated with the green transition has been recognized by the EU. One of the central cornerstones of the European Green Deal is the approach of ‘just transition’: the idea that the green transition is to be implemented in a ‘just’ way, without increasing inequalities among European citizens:

“Citizens, depending on their social and geographic circumstances, will be affected in different ways. Not all Member States, regions and cities start the transition from the same point or have the same capacity to respond. These challenges require a strong policy response at all levels”
(COM/2019/640, sec. 2.2.1).

Taking the above arguments into account, the proposed project will shed light on the potentially unequal effects of the sustainable structural reforms in the EU on different countries on the national, regional as well as local levels. This paves the way for the delineation of the multi-level policy measures that the Commission mentions in the Green New Deal, both to prevent asymmetric effects and to allow for an inclusive transformation towards a more sustainable Europe.

1.2. Research questions and hypotheses

Work package 1

Overall guiding question: *Does the green transition in Europe fuels socio-economic polarization among EU Member States, at least in the absence of suitable counter-measures?*

Concrete hypotheses:

- H1: The reforms that are required to put the EU on an emission reduction pathway consistent with the Paris Agreement are associated with heterogeneous transition costs across countries.
- H2: The costs are systematically higher for countries that are currently catching-up in terms of GDP and technological capabilities; thus, their economic development model is threatened.
- H3: The capabilities to engage in environmentally sustainable activities are unequally distributed across the EU Member States, with the Member States in Central Europe (the ‘core countries’, see below) having an advantage over the rest of the European Union.

Work package 2

Overall guiding question: *To what extent are EU coal regions prepared for the implementation of the European green energy transition policy?*

Concrete hypotheses:

- H1: Coal regions with suitable geographical conditions for the introduction of renewable energy sources will suffer smaller transition costs during an energy transformation
- H2: Coal regions with more related industries (as measured by technological and skill relatedness) will suffer smaller transition costs during an energy transformation
- H3: High intensity of eco-innovation in the field of energy in coal regions increases the chances of a successful green energy transformation
- H4: Coal regions differ with regard to the energy intensity of their related industries; this is relevant since high energy-intensity of industry in coal regions constitutes significant constraints in the implementation of the green energy transformation policy

When testing the hypotheses, the work package will compare coal regions among each other, and coal regions to non-coal regions.

Work package 3

Overall guiding question: *What are the differences and similarities in energy transformation processes between Konin (Poland) and Lusatia (Germany), and their positive and negative outcomes for the local communities, the environment, and economy in relation to the European Just Transition framework?*

Concrete hypotheses:

- H1: The infrastructure adapted to lignite with the current production and consumption patterns of energy in both regions (Konin and Lusatia) is the greatest obstacle to the energy transformation, and the funds allocated to it asymmetrically affect the transformation of both regions.
- H2: Local energy consumption, with the regions’ average attractiveness in terms of RES energy production, forces the involvement of all local entities in the green transformation led by the companies LEAG (Lusatia) and ZE PAK (Konin) - the local producers of electricity obtained from lignite.
- H3: Even though the capabilities for environmental activities, including technological potentials and skills, are different for both regions (with Lusatia being more advanced), the chances for a green transformation are the same, especially due to ZE PAK's investments in green hydrogen - in line with the assumed transformation strategies (Konin by 2030 and Lusatia by 2038).
- H4: Activities in the field of environmental innovation, including Konin's focus on the production of hydrogen fuel, as well as national governments and EU investments, allowed both regions to implement their plans for the development of green energy. At the same time, they led to the

deepening of social inequalities (social costs) resulting from the different set of local capabilities needed for sustainable activities.¹

Work package 4

This work package is geared towards the identification of vital research linkages between the three levels of analysis as well as at formulating policy recommendations. Thus, it does not feature concrete hypotheses to be tested, but rather guiding questions meant to facilitate the synthesis of the previous results

Guiding questions:

- Q1: Did the Study on the national level find evidence for ‘macro foundations for micro’ (King 2012), i.e. top-down effects such that specific macroeconomic developments constrain the behavioral and policy space of the actors on the regional and local levels?
- Q2: Did the explorative research on the local and regional levels suggest new foci of analysis on the macroeconomic level in terms of bottom-up effects for the green transition? More precisely, was there evidence for micro-scaling phenomena, like the actor-network-theory (ANT, Latour 2005)?
- Q3: Do the insights from case studies on local and regional level for Germany and Poland align well with the placement of these countries during Study 1? Or do the results suggest ‘fractal’ classification of countries and their regions, where differences of regions between countries are not necessarily larger than differences within the same country?
- Q4: What is the main understanding of a ‘just transition’ as aspired by the European Green Deal from the viewpoint of the local and regional actors examined throughout Studies 2 and 3? What are the main conflicting interests and viewpoints?
- Q5: For the macroeconomic levels, some politicians advanced the idea of a ‘multi-speed Europe’ according to which some EU countries will cooperate more closely with each other in certain areas than with other countries. How does this proposal align with within- and across-country heterogeneity on the national, local and regional levels identified before?
- Q6: What policy areas should be concerned to counteract potentially asymmetric effects of a green transition on the various levels, and what amount of funds would be necessary to effectuate them?

2. Significance of the project

2.1. State of the art

The investigation of the socio-economic effects of the necessary socio-ecological transformation of the EU requires a consideration of distinct branches of literature. This project builds directly on the insights of previous studies dealing with economic consequences of sociotechnical transitions found in transition studies (2.1.1), the economics of socio-economic polarization in Europe (2.1.2), the inquiries into local ecological transformation by regional studies (2.1.3), as well as the specific literature on the two regions that serve as representative case studies in this project: Konin (Poland) and Lusatia (Germany, 2.1.4).

2.1.1. Multi-level nature of energy transitions

The literature recognizes that **energy transition towards renewable energy sources** is not merely a technical issue, but one **spanning social, political, technological and economic factors** (Fouquet & Pearson 2012; Grubler 2012). Empirical studies on the historical events of sociotechnical transition demonstrate their large-scale and multidimensional consequences, especially for the functioning of diverse economic sectors and for the standards of living of societies (cf. Geels 2011). It spans transformative processes at different levels of socioeconomic systems, from the microeconomic level (e.g. technologies and production organization in businesses, consumer practices), through regional and sectoral (e.g. dominant patterns of accumulation and specialization, diverse types of competitive advantages), to the macroeconomic level (institutions, systems of innovation and overall competitiveness). In turn, the actual efficacy of transition depends largely on the involvement of diverse actors and the interplay of their political and economic interests.

Transitions in the energy sector are especially profound due to the vital role of energy sources, fuels and infrastructure for the functioning of all other sectors of the economy, as well as for the society at large. Moving away from traditional, coal-based energy provision invokes a ‘regime destabilization’, and often bears the danger for political conflicts with the incumbent mining and energy businesses - as evidenced by the recent cases of Germany and South Africa (Leipprand & Flachslund 2018). Incumbent companies, but also their employees and the whole regions, are most vulnerable to the radical change in energy policy. Historical evidence demonstrates that coal phase-out has had a long-standing negative impact on the affected regions,

¹ These include access to technology, construction of infrastructure, access to capital, RES education, employment structure, political will, as well as restoration/revegetation strategies and development of the region.

with prolonged undermined employment, business activity and standards of living (Spencer et al. 2018). In turn, public policies become vital, with their potential both to mitigate the negative effects of coal phase-out, but also to ignite new technological and economic activity. In this context, the advantages of a multi-scalar approach that span local, regional and national factors are being recognized both in research and in policies (Pegels & Lütkenhorst 2014; Oei, Brauers & Herpich 2019). Moreover, regional studies highlight the diverse aspects of scaling and spatial nature of energy transitions, including the geographies of infrastructures and knowledge, and the weight of national policies and institutions (Dahlmann, Kolk & Lindeque 2017; Bridge 2018). **The present project** takes this new consensus on the multi-level nature of the problem of energy transition as a starting point and provides for a novel application of the concept within the so far understudied context of the European green transition.

2.1.2. *Structural polarization and different growth models in the EU*

The proposed project draws from the existing literature on the **political economy of European development**, especially that concerned with growth models and supply chain integration within the European Union. The literature on **growth models** seeks to identify the main drivers for economic growth in a given economy on the demand side (e.g. Baccaro & Pontusson 2016; Gräbner et al. 2020a). Today, the **main growth models** currently present in the European Union are the **export-led growth model**, as pursued by countries such as Germany or Austria, and the **debt-led growth model**, in which aggregate demand gets stabilized by mainly domestic actors who are willing to incur debt in order to boost demand (Gräbner et al. 2020a). This latter strategy, which has mainly been pursued by countries in the South, has proven to be unsustainable during the financial crisis, leaving these countries in an economically detrimental position, as compared to those countries following an export-led growth model (Gräbner et al. 2020a, 2020b). Most countries in Eastern Europe take the special position of **catch-up economies**, who currently suffer from rather low per capita income, but are catching-up in terms of productivity, capabilities, and income. Whether they will be able to converge, however, to the level of Central European core countries, however, remains to be seen and depends, inter alia, on the institutional construction of the European green transition, as will be argued below.

Countries can only follow an export-led growth model if their firms are **sufficiently competitive** on the international markets. The main sources for such competitiveness could either lie in low factor costs rooted in low labor standards, limited regulation, and cheap inputs, or in high product quality, complexity, or specific product properties. Since the path of low regulations and low working standards is essentially ‘blocked’ by EU legislation, it is only a sufficient level of technological capabilities or low factor costs if combined with an integration into complex supply chains, that allow countries to follow an export-led growth model in the EU (Kapeller et al. 2018; Gräbner et al. 2020b). **Diverse capacity to meet the requirements of international competition** has led to a **structural polarization** among the EU member states, which affects the opportunity space for regional development policies today. This is particularly the case since the accumulation of new technological capabilities is a highly path dependent process (Aistleitner et al. 2021). Energy transition may add another risk factor to the polarization processes, as identified in the European comparative political economy (Ćetković & Buzogány 2016).

The literature on **global supply chains** complements this work and highlights the particular role of Eastern European countries: European manufacturing supply chains can be shown to consist of **‘headquarter’ and ‘factory’ economies**. Eastern Europe is largely positioned in the latter group (Grodzicki & Geodecki 2016; Grodzicki & Skrzypek 2020), providing mainly intermediate products for the high-complexity industries in Central Europe.² This is relevant for the future-fit renewable energy technologies: Lachapelle et al. (2016), for instance, demonstrate the emergence of a *Green Division of Labor*, consisting of leading patent innovators (e.g. Israel, Switzerland, Finland); manufacturers (e.g. Taiwan, Malaysia, India); and mere installers of technology (e.g. Czech Republic, Greece, Belgium). Notably, Germany achieved a leading position in several segments of the renewable energy technologies, which may mitigate the overall costs of the transition of its fossil-dependent economy. Poland, however, lags behind in terms of environmental innovativeness, in line with its general peripheral mode in division of labor. This may reinforce the estimated macroeconomic costs of coal phase-out in this country (Baran et al. 2020), indicating that a comparative analysis of the two countries is indeed promising, and that the challenges for regional policies differ systematically across countries.

The present project is innovative in the sense that it brings together these complementary branches of literature, and adds the so far neglected role of ecological sustainability and ecological reforms to the analysis of structural polarization.

² In fact, Eastern countries have substituted many of the Southern economies as main suppliers of the complex manufacturing sectors of Central Europe, one important reason for the failure of the Southern periphery to sustain an export-led growth model (Gräbner et al. 2020a).

2.1.3. Regional dimension of green transformation

The asymmetric effects highlighted by the literature on polarization in Europe point to the **regional dimension of the polarization challenge**: within countries, the dependency on environmentally harmful activities and the local set of capabilities needed for sustainable activities differ considerably both between and within countries (Ganzleben & Kazmierczak 2020; Bouzarovski & Tirado Herrero 2017). Moreover, the regional innovation systems as well as the institutional framework matter since local and regional authorities play an important role in creating and implementing green policies and the legal and regulatory framework for energy transition (Bush 2020; Grillitsch & Hansen 2019).

An energy transition as envisaged in the Green Deal is a source of risk and opportunities for coal regions: while energy poverty and the need to adjust to the new market structure are a main source of vulnerability (Dudek & Szlagowski-Budacz 2019), the renewable and clean energy sector also bears many opportunities, such as the chance to create new jobs (Kapetaki 2020; Alves Dias et al. 2018). Thus, given the need to phase-out coal production in order to reach a 1.5 degree path, the EU coal regions are likely to experience **considerable costs of a green transformation**, at least in the short run. Whether and when the development of the green energy sector can become an opportunity for new jobs creation depends on specific institutional factors. The second work package will identify these and assess their socioeconomic implications, thereby providing for the novel contribution of studying regional effects within a broader multi-scale investigation of the European green transition.

2.1.4. Lignite in Germany and Poland, with emphasis on Konin and Lusatia

To put their economies on a 1.5 degree-compatible emission reduction pathway, both Germany and Poland must take radical steps to move away from fossil fuels, especially **lignite**, which causes **particularly high emissions** to the environment (not only CO₂ but also other toxic pollutants, EEA 2019). At the same time its **calorific value is much lower** than one of hard coal or natural gas (Umwelt Bundesamt 2016). Nevertheless, it remains essential in the energy-mixes of both – Polish and German – economies, constituting **almost 20% of the countries' energy generating capacities** (AGEB 2020). Lignite mining and energy complexes are **highly concentrated geographically**, which means that the coal phase-out is likely to constitute a large asymmetric shock for particular regions. The present projects takes two such lignite regions – Konin in Poland and Lusatia in Germany – as case studies.

The qualitative research conducted so far in mining regions, especially concerning the extraction of lignite, has mostly focused on the socio-environmental aspects of exploitation (cf. experiences from central Poland, Badera & Kocoń 2014), interviews with inhabitants of the destroyed villages of Lusatia, Morton & Müller 2016) or on waste utilization/management (cf. waste from the mining of rock raw materials in Lower Silesia, Kazmierczak et al. 2018). However, so far, **no fieldwork research was carried out either in Konin or in Lusatia concerning the ongoing energy transformation**. Thus, the present project facilitates research that is innovative by (1) illuminating the relevance of macroeconomic constraints discussed above, (2) highlighting the relevance of informal (and often local) institutions that are difficult to consider in studies targeting higher scales, and (3) allowing to study and map the transformation taking place in both regions, including investments in innovations and research (resulting from - among others - works on **hydrogen fuel** in Konin).

2.2. Justification for tackling a specific scientific problem

The literature review points to a number of relevant **research gaps**, which the project is meant to address:

1. The topic of energy transition is timely and highly relevant both from a theoretical and a policy view; new empirical results to guide policies are greatly needed;
2. The relevance of macroeconomic constraints and local (potentially informal) institutions for regional policy has so far not been studied within a consistent multi-level framework;
3. The triangulation of quantitative and qualitative research methods under a common mixed-methods approach is particularly well suited to study the impact of EU transition policies, yet remains underexplored in the existing literature;
4. The consideration of the necessary policy measures to comply with international treaties is a blind spot of the literature of economic polarization and growth models in Europe; at the same time, the literature on the green transition has so far neglected the impact on socioeconomic polarization and the relevance of growth models, despite clear linkages between the subjects;
5. The economic linkages between energy transition, inequalities, and structural disparities have received insufficient attention so far (as indicated in Chester 2014).

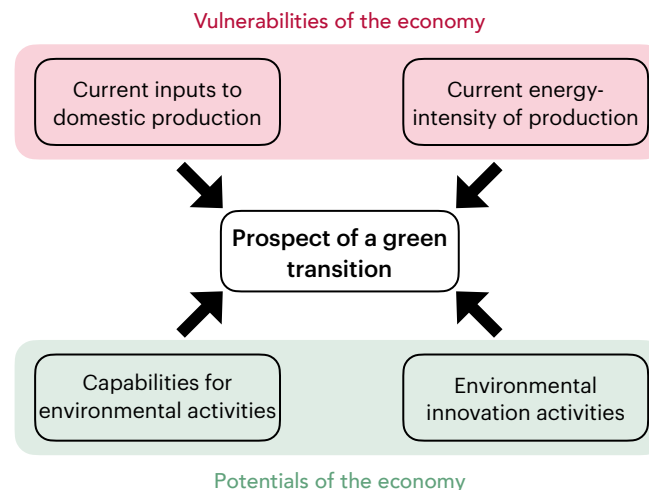
These gaps clearly justify the essence of undertaking the presented research problem of a just and green energy transformation in Europe.

2.3. Justification for the pioneering nature of the project

A central innovation of the proposed project is that, first, it investigates the impacts of the green transition on these different scales of the economy in **one consistent framework**, and, second, takes a **mixed method approach** considering both quantitative and qualitative sources of information.

The **consistent framework** for the analysis consists of (1) an explication of the **main substantial areas** covered in the proposal, and (2) an explication of the **interrelation between the local, regional, and national** levels of analysis within the meta-theoretical framework of ‘systemism’ (Gräbner & Kapeller, 2015). From a substantial perspective, the project is based on the idea that the **overall prospect of the European green transition of a country** (or a region) depends (i) negatively on its specific **vulnerabilities** and (ii) positively on its specific **potentials**. Regarding the vulnerabilities, the present project focuses on (i-a) the current energy-mix and set of input factors of an economy and (i-b) on its current economic specialization as measured by the composition of its products and services. When it comes to the specific potentials of an economy, the focus is on (ii-a) its innovation activities and (ii-b) the existing set of technological capabilities accumulated in the economy under consideration. Figure 1 summarizes these substantive pillars of the project and highlights again that the overall effect of a green transition in any economy is determined by the combined effect of these four dimensions. The consistent application of the framework to all three levels of analysis will enable us to assess the actual role of structural and technological factors in these processes.

Fig. 1: The four dimensions that determine a country's prospect in a European green transition and that are used to guide further analysis.



The **mixed-methods approach** (Greene 2007) is innovative and promising since it allows taking seriously the local specificities and regional heterogeneities within the EU, as well as the relevance of **informal (and often local) institutions** for regional policy design. A particular feature of the mixed-methods approach in this context is its comparative view on Germany and Poland: this is in contrast to the latest research in Lusatia (e.g. Heer et al. 2020) discussing the guidelines for designing the transformation process, yet not examining the ongoing process per se. Moreover, the **comparative analysis** that will be carried out in this project will expand upon the insights of such single-case studies and allow them to make contact with the literature on comparative development in the EU.

2.4. The impact of the project results on the development of the research field and scientific discipline

The expected impact of this project on the development of science is to shed light on the potentially unequal effects of the sustainable structural reforms in the EU on the national, regional as well as local levels, and to delineate a set of regional policy measures that prevent an increase in inequality and facilitate an inclusive transformation towards a more (socially as well as ecologically) sustainable Europe. Therefore, this project is both academically attractive and socially highly relevant.

The project contributes to the field of regional studies via an explicit **multi-level research design**, as postulated in the geographies of energy transition literature (Dahlmann, Kolk & Lindeque 2017; Bridge 2018). The **analytic benefits of such a multi-level analysis** are at least twofold: first, it allows us to **take into account the heterogeneity on each of the three levels** when it comes to the four dimensions (Fig 1): in Germany, for instance, the energy-intensity of production is extremely high in the Ruhr area, due to the regional specialization on the steel industry. The Rhein-Main area, on the other hand, is focused on the provision of complex services, which require far less energy as an input source. Neglecting this heterogeneity

bears the danger of misleading conclusions. Therefore, Studies 2 and 3 explicitly take into account this heterogeneity.

Moreover, the multi-level approach allows one to **take seriously macroeconomic constraints for regional developments**: focusing on the regional level alone bears the danger of omitting important national factors that themselves feed back into the development on the regional and local level. The specializations of Germany and Poland, for instance, are such that firms in the automotive sector in Germany benefit from considerable economies of scope and the considerable stock of technological capabilities in this area, while in Poland such benefits materialize more in the area of labor-intensive processes via lower factor costs.

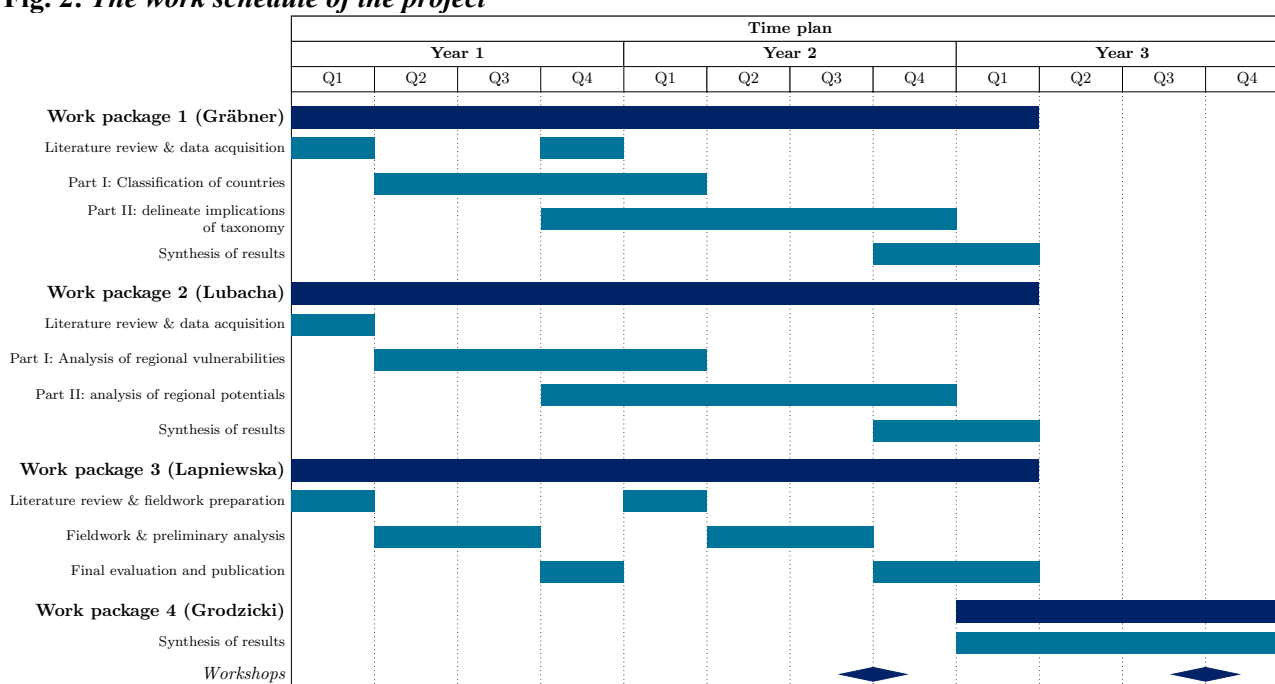
The project also contributes to the field of regional studies via an **innovative triangulation of distinct methods** that pay tribute to the heterogeneity of mechanisms and actors operating on each level of analysis. While the main ‘actors’ on the national level are European member states, the meso level focuses on regional conditions and capabilities for energy production from RES, and the local level puts specific firms, local governments and communities at the centre of analysis. And while tools such as input-output modelling or macroeconometric analysis are useful to identify structural determinants for the development paths of whole countries or regions (which constrain the economic actions and policies that are available on the regional and local levels), they are less helpful in identifying local capacity to change economic routines towards more sustainable activity. Here, interviews are the more promising route to take. Our project combines the respective strengths of these distinct methodologies within a consistent theoretical framework.

3. Research concept and work plan

3.1. General work plan

The project consists of four work packages. Work packages 1-3 will be carried out simultaneously by Polish-German teams. The fourth work package closes the project, synthesizing previous results, and will, therefore, be a common endeavour for all collaborators. Figure 2 summarizes the resulting time schedule.

Fig. 2: The work schedule of the project



3.2. Specific research goals

Work package 1 (national level): the national-level constraints for regional development

Leader: Prof. Claudius Gräbner-Radkowsch Collaborators: Dr. Maciej Grodzicki, PhD Student

Time Span: 01/2023-03/2025 (EUF)

Consultants: Prof. Jakob Kapeller (Duisburg-Essen), Prof. Till van Treeck (Duisburg-Essen), Prof. Ulrich Glassmann (EUF)

The first work package is meant to identify **macroeconomic constraints and implications for regional policies**. The guiding question is whether – given the current institutional framework in the EU – the

implementation of the policy measures necessary to comply with the obligations as set out in the international treaties surveyed above impacts differently (or even asymmetrically) the EU member states. The work package will identify the implications for regional policy design, i.e. whether location-specific regional policies are needed to counter-acted potentially unequal effects of a green transition and how they would need to look like, or what the best general guidelines for regional policies would be if such unequal effect was not identified.

The work package builds upon the growth models framework (cf. section 2.1.2), adjusted for the energy transition context. The economically more successful EU member states from the macroeconomic perspective clearly follow an export-driven growth model (see also Gräbner et al. 2020a). Therefore, their **growth potential depends on certain competitive advantages**, i.e. low costs or high product quality and sophistication. The **need to transform their economies** towards more sustainability might **put the successful pursuit of an export-led growth model at risk** by eliminating their current competitive advantage: countries such as Poland, for instance, rely on the extraction of exhaustible resources such as coal, and have not yet accumulated the technological capabilities required to produce complex, but environmentally less harmful goods. For countries such as Germany, not only are the structural reforms relatively less demanding, their economy is also more diversified, easing the transition towards the production of alternative complex products. This state of affairs bears the danger of polarization.

More concretely, the study aims at delineating a country taxonomy along four dimensions, each one representing one of the main dimensions of the project as illustrated in Figure 1. In the **first dimension** countries are classified according to the **main inputs used by their domestic economy** and the existing sources for domestic energy production. The **second dimension** is concerned with the energy intensity of the production itself, in particular the export-oriented economic activities. Here, one might expect **countries that are currently specialized on relatively energy-intensive activities will face greater costs and challenges** than others and are likely to lose ground in terms of economic development. The **third dimension** takes into consideration the **innovation activities** of the member states in the environment-related technologies. Environment-related technologies, especially in the energy generation and other mitigation activities, are likely to play a more important role in international trade. Thus, ensuring competitiveness in the production of these technologies is an important determinant for lasting export success and the feasibility of export-driven growth. The **final dimension** focuses on **actual production and service activities** (in contrast to R&D activities considered previously) and considers the existing set of technological capabilities in the countries. Combining this information with data from environmental input-output tables will allow the identification of those countries that have already accumulated capabilities needed for environmental-friendly activities, and those that are focused on activities that are rather unrelated to environmental-friendly activities. Since countries can only develop towards related activities (see Hidalgo et al. 2018), the distance of their current specializations to environmentally-friendly activities gives a good indication of how considerable are the costs of transition, and how much the country can benefit from the increased relevance of environmentally-friendly products and services.

The work package then builds upon the country classification and study the **implications for comparative economic development in Europe**, as well as **the policy and action space for regional actors**. To this end we relate the new classification to existing taxonomies for development models, especially the one proposed in Gräbner et al. (2020a), who have identified relatively well-off core countries ('core countries'), an economically weaker European periphery and a group of currently poorer countries that are on the pace of catching-up to richer countries. The latter group mostly consists of countries in Eastern Europe. The present project will further investigate whether the need for a green transition tends to reinforce these country groups and whether it poses a threat to the catch-up dynamics in Eastern countries.

Work package 2 (regional level): A Comparative Analysis of the European Coal Regions

Leader: Dr. Judyta Lubacha

Collaborators: Dr. Zofia Łapniewska, Prof. Claudius

Time Span: 01/2023-03/2025

Gräbner-Radkowsch, PhD student (Poland)

Consultants: Prof. Andreas Pyka (Stuttgart-Hohenheim), Dr. Tobias Wendler (Bremen); Prof. Maciej Smętkowski (Warsaw)

To get a clearer view on the relevance of the just mentioned risks and opportunities, the second work package employs a regional level analysis, which is expected to provide a wider picture of transition capabilities in European coal regions. In 2020, the European Commission launched the Just Transition Mechanism, which was meant to support the transition towards a climate-neutral economy and to address its socioeconomic consequences (COM (2020) 22 final/2020/0006 (COD)). The decisive question for the coal regions in the European Union is whether they are able to exploit these potential benefits, i.e. whether they already have, or

are able to accumulate the necessary level capabilities to switch from coal mining and coal-based energy production to innovative clean production sectors. In this context, the regional preconditions toward green energy transition will be analyzed regarding four dimensions: first, geographical conditions for the introduction of renewable energy sources, second the presence of linkages to related industries (as measured by technological and skill relatedness), third, the intensity of eco-innovation in the field of energy in coal regions, and, fourth, with regard to the energy intensity of their related industries, the latter issue being relevant since high energy-intensity of industry in coal regions constitutes significant constraints in the implementation of the green energy transformation policy. In all cases, both formal and informal institutions will be considered. Formal institutions will be explored based on information from regional and national level strategies and regulations, while the informal institutions will be derived from the analysis of social norms and attitudes toward energy transition. Moreover, the economic structure of the region will be pictured with focus on sectors operating in a region, innovation activity and innovative capabilities, as well as the present utilization of green energy sources.

Work package 3 (local level): Comparative Fieldwork Research of Two Lignite Mining Regions - Konin (Poland) and Lusatia (Germany)

Leader: Dr. Zofia Łapniewska

Collaborators: Dr. Judyta Lubacha, PhD student (EUF)

Time Span: 01/2023-03/2025

Consultants: Dr. Özgür Yildiz (TU Berlin), Dr Hab. BS. Krzysztof Berbek (Krakow), Prof. Pao-Yu Oei (EUF)

This work package comprises a qualitative analysis of the **impact of the energy transition at the local level** in the scope of a **comparative fieldwork study** of two lignite regions - **Konin (Poland)** and **Lusatia (Germany)** - to answer the main research question and verify four hypotheses (see point 1.2 of this proposal). In both regions mines as well as lignite-fired power plants are owned by private companies, especially ZE PAK S.A. (Konin) and LEAG (Lusatia). On October 1, 2020, the ZE PAK company announced that it would withdraw from lignite-based energy production by 2030, being “the first large energy producer in Poland to produce all its electricity from renewable sources” - as the owner of the majority stake of the company, Zygmunt Solorz-Żak, announced (ZE PAK 2020). A year later, the company received a grant from the European Union Innovation Fund - in the amount of approx. EUR 4.5 million - co-financing the construction of a green hydrogen production plant (ZE PAK 2021). Doubts regarding the feasibility of the envisaged target presented by the media relate mainly to the appropriate selection of technologies and the potential for retraining and employment of professionals operating these technologies. At the same time, Germany adopted a coal exit law in July 2021, according to which the whole coal-fired power generation is meant to phase out by 2030. The law also includes compensation payments for energy companies and financial aid for coal mining regions, incl. Lusatia (BMW 2020). While ecological NGOs criticize the law for its allegedly low ambition and the late phase-out date, others criticize the high cost of phasing out from coal, especially in the era of COVID-19. For instance, the ‘Black Book’ report, published by The German Taxpayers Federation on October 28, 2020 criticized the multi-billion euro plan, pointing out that the EU Emissions Trading System would lead to unprofitability of the coal-fired power plants anyway and eventually contribute to their closure (DG Climate Action 2015, Meza 2020).

Recent research on these two regions (i.a. Szpor & Kiewra 2018, Kasztelewicz 2018, Graichen & Rosenkranz 2018) points out the **lack of empirical research at the micro level illustrating the economic complexity of the energy transformation for these regions**. Given the lack of relevant data on the local level, this Study will engage in fieldwork research of a primary nature, geared towards an actual diagnosis of the economic situation (including social and environmental conditions), a critical verification of existing plans and strategies, and a comparative institutional analysis of two lignite basins that have a high relevance for the Polish and German energy markets.

Work package 4 (synthesis): Implications for future research, regional policies, and the implementation of the European Green Deal

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Time Span: 01/2025-12/2025

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The final work package is meant to, first, **synthesize the results of the three research Studies** conducted at the national, regional and local levels and, second, to **delineate the implications for the multi-level policy program** anticipated by the European Commission.

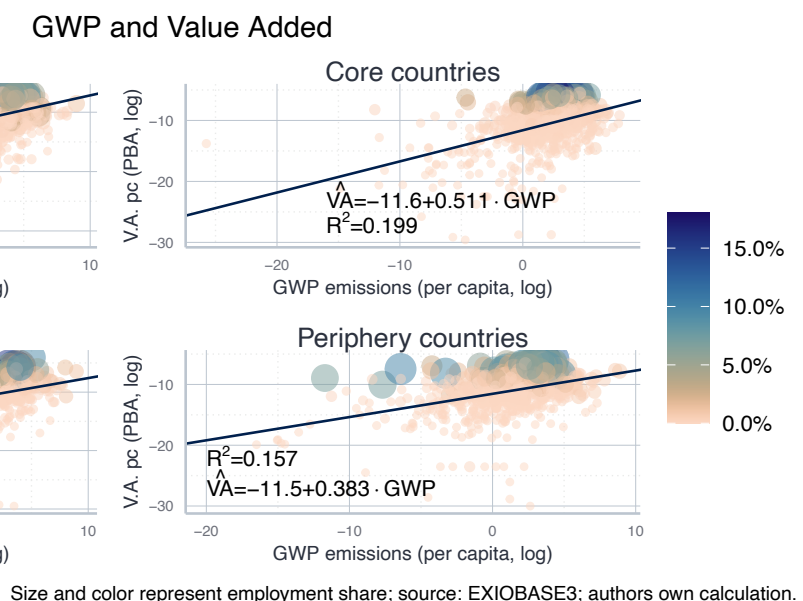
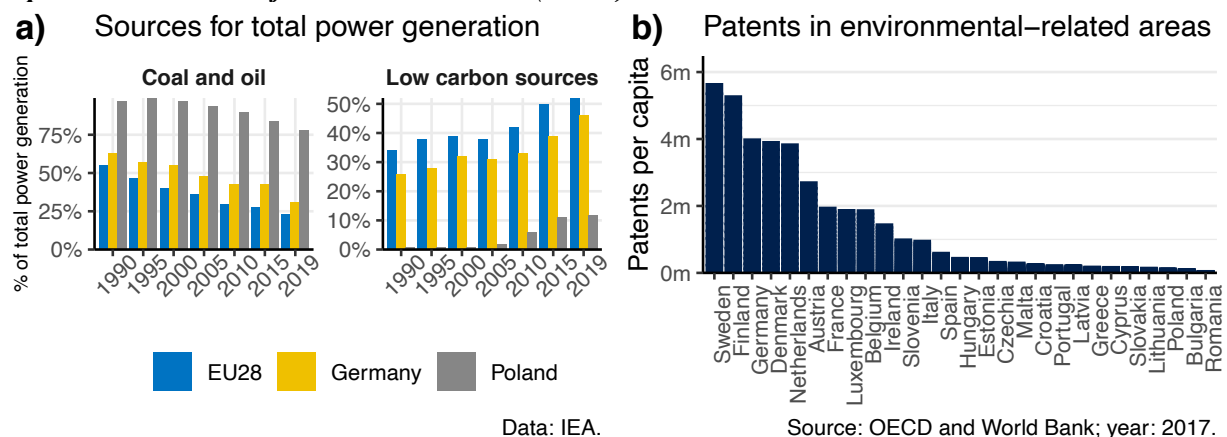
The synthesis of the results of all three Studies will be analytically embedded into the overall structure of the project, i.e. the four dimensions outlined in Figure 1 and the framework introduced in section 2.3. When interpreting the results, we follow the philosophical framework of systemism as described in Gräbner & Kapeller (2015). This framework allows for a clear distinction between the different levels with its specific actors and mechanisms, but also an explanation of ‘bridging mechanisms’ between different levels, i.e. mechanisms operating from the local to the national level, or vice versa. While the exact execution of this work package depends on the particular results of the first three packages, at this point six guiding questions that help to structure the final synthesis of results are delineated (see section 1).

After this initial synthesis, the implications of the project findings for the ‘strong policy response at all levels’ suggested by the European Commission (COM/2019/640, sec. 2.2.1) will be delineated. More specifically, it will be discussed how policy measures on distinct levels of analysis can take into account the heterogeneity in the reactions of different countries and regions to the green transition that can be expected based on the results from the first three Studies, and how such policies can be developed within the current institutional framework of the European Union.

3.3. Results of preliminary research

Figure 3a shows the unequal reliance on environmentally harmful energy sources for Germany, Poland and other EU countries. The data clearly points to an asymmetry, with Poland being much more dependent on fossil fuels, indicating that the transition costs for putting Poland on a 1.5 degree-compatible emission reduction pathway are much more severe than, for instance, for Germany. Figure 3b provides complementary insights by showing the per capita amount of patents in environmentally friendly areas: again, the data point to a very unequal distribution of patent activities, hinting at an unequal distribution of capabilities in these areas. Thus, countries such as Sweden and Germany are much more likely to benefit from a green transition than, for instance, Poland or Greece.

Figure 3: Preliminary results on the different sectoral specialization of the EU economies. The country groups are the one identified in Gräbner et al. (2020b).



Finally, Figure 3c shows the results of a regression with the total value added of a sector as the dependent variable, and the global warming potential of this sector as an explanatory variable. The regressions are carried out for the four country groups identified by Gräbner et al. (2020b). One clearly sees that the relationship is most pronounced for the group of catch-up economies, both in terms of the size of the relationship as well as the variance explained, although the former is of comparable size for the core countries. Thus, the Eastern catch-up countries are dependent on environmentally harmful activities to a much greater extent than, for instance, the highly financialized or periphery countries. The same holds true if one considers employment opportunities instead of value added (not shown in the figure).

In all, while these preliminary results need to be refined methodologically, and only provide for a very narrow view on the status quo, they do indeed suggest that challenges and potential benefits of a green transition are distributed very unequally, and that policies, especially on the region level, are necessary to overcome these inequalities and to prevent an accelerating polarization in consequence of the (necessary) green transition in Europe.

3.4. Risk analysis

Given the exploratory character of work package 1, only marginal risks exist. If, for instance, the country taxonomy to be delineated in the first study does, against our expectations, not fit to the one of Gräbner et al. (2020b), then this result would be interesting in itself and not represent any danger for the overall project. Thus, the following risks focus on Studies 2-4 and general project management.

Risk	Likelihood	Risk mitigation
No consistent fit between the typology of Study 1 and the other results	Unlikely	Depending on the identified topology, a theoretical rationalization of the misfit between national, regional and local economic dynamics would need to be developed. This would, however, not endanger the conduction of the other studies as such, but only aggravate overall interpretations.
Difficulty in addressing respondents in Study 3	Possible	To conduct the interviews the research teams will contact local authorities in advance as well as non-governmental organizations dealing with ecological topics to take advantage of their suggestions. Formal channels of communication with coal companies management will also be used in advance.
Underestimation of time or resources generating delay or affecting quality of work	Unlikely	Time schedules of all Studies and project expenditures will be carefully monitored over the 3-year period. Early indicators signaling that progress is falling behind schedule or expenditure is approaching budget limits will be discussed by the team, determining actions to address.
Managing workload problems and conflicting priorities	Possible	Detailed division of tasks between members of the project team will be defined at the kick-off meeting to ensure that the workload is spread across the team members adequately and knowledge shared. Monthly meetings of the project entire team (Polish-German) will ensure that there is clear communication of all issues and appropriate actions will be taken.

4. Research methodology

Work package 1 delineates a country taxonomy along four dimensions, each one representing one of the main dimensions of the project as illustrated in Figure 1. In each dimension descriptive statistics, as well as methods from the field of unsupervised machine learning, such as principal component analysis and clustering to uncover an underlying typology of countries, will be used. Information about sectorial specializations, different degrees of integration into supply chains, and the environmental footprints (both according to a production-based and a consumption-based accounting approach, see Tukker et al. 2020) will be obtained via tools from input-output modelling. Their relation to other important economic indicators (such as employment or value added) will be established via regression analysis (see the preliminary results in Section 3.3). When analysing patents in the third study we will also use network analysis to uncover the relatedness structure underlying the

patents. In the fourth dimension, which considers the actual production and service activities, the method to compute indicators of economic complexity as suggested by Hidalgo & Hausmann (2009), and the concept of the product space as initially proposed in Hidalgo et al. (2007) will be applied. Finally, a bridge between study 1 and 2 could be built using the tools of Mewes & Broekel (2020), who compute complexity measures for regions based on patent data. Data on energy intensity and energy use is provided via Eurostat and the EU Klems database. Environmental footprints and the supply chain analysis rely on input output tables, which are provided by the OECD, as well as Timmer et al. (2018, WIOD) and Stadler et al. (2018, EXIOBASE). Patent data are taken from the OECD and the PATSTAT database, and export data from the Observatory of Economic Complexity.

Work package 2 delineates a regional taxonomy along four dimensions, each one representing one of the main dimensions of the project as illustrated in Figure 1. The subject of the analysis are 21 EU coal regions. In each dimension we will use descriptive statistics, as well as clustering to uncover an underlying typology of regions. In addition, geostatistics (Webster & Oliver, 2007) will be used to present the spatial dimension of described phenomena. Structural specialisation of the regions will be analysed based on Eurostat regional statistics (business statistics, employment). Because of the lack of standardised regional input-output tables, the possibility to describe integration of the coal regions into supply chains is limited. Regional data about energy intensity and energy consumption has to be collected from National Statistical Offices since they are not provided by Eurostat at NUTS2 level. Environmental footprint data on a regional level will be obtained from experimentally developed datasets (eg. Mura et al., 2021; Ivanova et al., 2017). Current energy production and potential for use of renewable energy sources will be supported with GIS data analysis of solar irradiation and wind speed. Environmental innovation activity will be pictured based on mapping eco-innovations (patent data) developed in an energy sector (Griliches 1990). Using tools proposed by Mewes & Broekel (2020), the technological complexity of regions and the relatedness of coal-(in)dependent industries will be measured. Finally, formal (regional and national regulation) and informal (attitudes towards energy transition based on European Value Study) institutional framework towards energy transition will be described.

Work package 3 comprises fieldwork research. Qualitative research – as a complementary part of the entire project – is necessary to get a comprehensive view on the subject matter. The first stage of this work package will focus on the analysis of data from secondary sources. In the second stage, data from primary sources (interviews and additional documents) will be collected as a result of fieldwork research. The final stage will focus on the synthesis of all acquired data (using the ATLAS.ti software).

The work package follows a mixed methods approach. The research design uses both quantitative and qualitative data (Hanson, Creswell, Plano Clark, Petska & Creswell 2005). Quantitative methods will be applied to regional economic data to scrutinize the transition processes in the two regions. Sources of secondary data will include, among others, statistical data, normative acts, documents of the councils and companies (ZE PAK and LEAG), documents created by the councils and firms, data sets and the mass media. For specific data collection during the first phase of this work package (preparation to fieldwork research), a rigorous research design will be developed, including a study flow diagram with research questions for normative acts, a questionnaire for document analysis, a questionnaire for data sets and research questions for mass media analysis (Stoker & Evans 2016). Further meaning to the analyzed secondary data will be added by a qualitative follow-up data collection (Johnson & Onwuegbuzie 2004; Łapniewska 2017, 2019a, 2019b).

Up to ten in-depth interviews will be conducted in every location each year (2023 at first and in 2024 follow up; Hesse-Biber & Leavy 2011: 93-130) with land and local governments' representatives, ZE PAK and LEAG managers, trade unions, local entrepreneurs, environmental organizations and researchers who design/analyze/forecast energy transition in Poland and Germany. Annual 5-day stays at both locations are included in the project budget. The interviews are going to be followed by observations (Hesse-Biber & Leavy 2011: 193-226) of the progress of the transition in Konin and Lusatian lignite mining areas. Additionally, we will establish Futures Literacy Labs (the UNESCO framework) and conduct group interviews/workshops for stronger exchange between stakeholders. Two sources of information will be used – personal (interviews) as well as direct and indirect (observations and measurements). The ATLAS.ti software will facilitate the process of organizing transcripts of interviews, fieldwork notes, observation sheets in addition to the quantitative data described above. These qualitative methods will enrich the analysis by contextualizing the selected places, giving meaning to specific citizens' needs and experiences (just transition), and finally allowing to use an inductive approach to understand, map and compare the observed processes between the regions.

The work package also follows the multiple case-studies analysis (Rihoux & Grimm 2006). This allows for a higher level of generalization and, at the same time, produces in-depth insight into the analyzed cases (Brady

& Collier 2004). The Qualitative Comparative Analysis method (QCA, Ragin & Rihoux 2004, Ryan 2014) is a statistical data analysis technique that indicates which logical conclusion supports a particular data set. It requires the collection of precise quantitative and qualitative data (Creswell 2014) for the definition and operationalization of variables that would examine different combinations of preconditions generating different (or similar) results (Ragin & Rihoux 2004, Rihoux & Grimm 2006). QCA will help in this Study to make appropriate use of sources and collected data, compare transition processes and generalize empirical findings. The ATLAS.ti software will allow to collect all the analyzed documents and not only makes them accessible for Polish and German team members, but also facilitates the process of managing the material in a systematic way. The data will be stored on computers purchased in the project, transferred using a password-protected USB Stick. To inform local communities and governments about the research in an accessible way, we will use the project website (in UJ domain).

Work package 4 comprises a synthesis of the research results, informed by the methodology of mixed-methods integration and triangulation (Fielding 2012; Heath 2015). We adopt a 'convergent parallel' research design, meaning that the three studies at diverse geographical scales are conducted at the same time, and only after the analyses are complete, we will compare the results to draw overall conclusions. We will assess the overall consistency of results at local, regional and national scales, as well as between qualitative and quantitative data sources. The synthesis will be oriented not only at validation of results, but rather at extending them and building a broader and systemic understanding of the relation between energy transition and economic polarization. In particular, we will search for vital complementarities (positive feedbacks), contradictions (negative feedbacks) and scaling mechanisms (bottom-up and top-down bridges) between the three studies – in line with the guiding questions Q1-Q3 (section 1.2).

In practice, we shall, first, build a 3-times-4-matrix representing three levels and four dimensions that determine a country's prospect in a European green transition. Comparisons across rows and columns of the matrix will enable us to identify the aforementioned complementarities and contradictions. This will be supported, second, by data aggregation and joint analysis by means of visual and statistical techniques (cf. Kadushin et al. 2008; Fielding 2012). The integrated results will inform the final process of formulating policy recommendations, which would support the just transition and help avoid asymmetric effects of energy transformation in the EU. In this process, the research conclusions from three studies shall be juxtaposed with the perspectives of local stakeholders and decision makers during the two workshops.

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